

X-ray imaging on Lead Carbonate

Timepix4 project

Linda Biasci

11/06/2026

Introduction

The Timepix4 detector is employed to perform X-ray imaging on a sample of Lead Carbonate (white pigment on wood) aiming to identify its K-edge and characterise the instrumental response and the possible limitations of such analytical technique. X-ray transmission of the studied sample has been previously analysed with an Amptek X-ray detector, resulting in a clear visualisation of lead's K-edge and allowing to estimate the pigment's thickness (assuming the wood's thickness to be known and neglecting the presence of a binder on wood).

Experimental setup

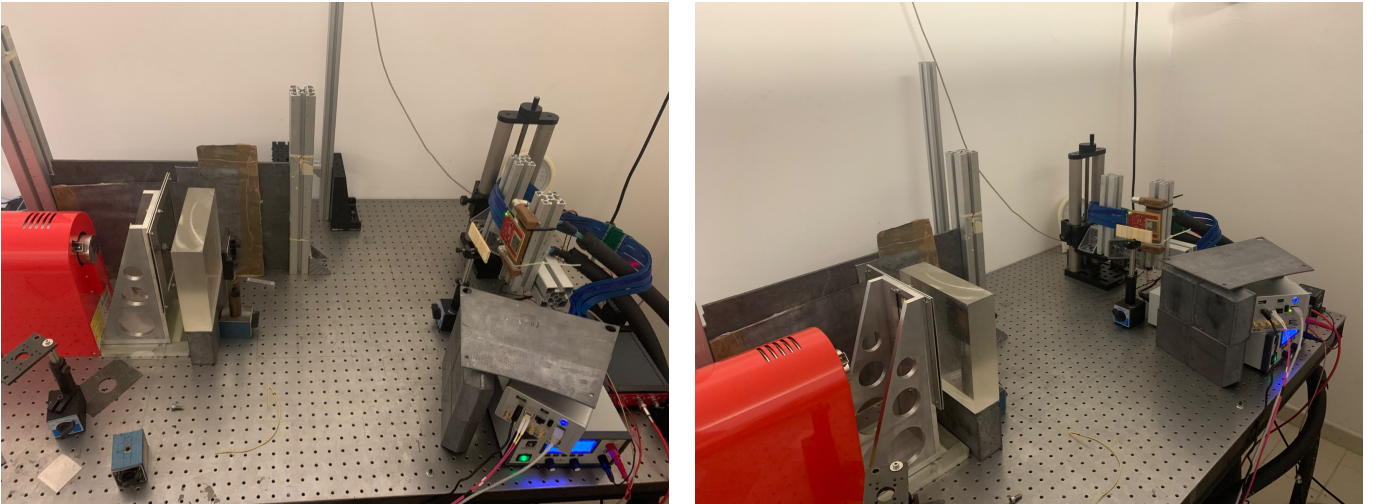


Figure 1: Experimental setup; from left to right: X-ray tube, PMMA attenuator, Al and Cu filtration, sample (white pigment on wood), Timepix4 in thermal contact with the heat exchanger and in electrical contact with the read-out system (shielded by lead plates)

The X-ray detector is placed behind the sample, at an approximate distance of 7 *cm*, and the distance between the source and the detector is approximately 80 *cm*.

Source

The employed source is a micro-focus X-ray tube (Hamamatsu), characterised by:

- tungsten anode in transmission mode
- energy range: $kVp = 120 \text{ keV}$
- anode current: $I = 10 \text{ }\mu\text{A}$
- filtration: an aluminum filter of width 1 *mm*, four copper filters of width 250 μm each
- attenuation: a PMMA attenuator of width $5.2 \pm 0.1 \text{ cm}$

- acquisition time: 10 runs, each with an exposition of 100 s (total acquisition time of 1000 s)
- collimator: a tungsten disk of thickness 2 mm, with a hole diameter of 1 mm

Sample

A Lead Carbonate pigment ($PbCO_3$) spread on a wood table; previously estimated thicknesses:

- wood: 5.036 mm
- Lead Carbonate: 10.096 μm

Detection system

Timepix4 is used with a CdTe detector, so that the detection system is characterised by:

- CdTe thickness: 1 mm
- expected energy resolution: $6 \text{ keV} < FWHM < 10 \text{ keV}$
- pixel area: $55 \times 55 \mu m^2$
- detector area: 448×512 pixels, i.e. total detector area of 6.94 cm^2
- cooling system: Timepix4 is placed in thermal contact with a copper heat exchanger and a chiller liquid continuously flows inside of it
- readout system: SPDR4 (Speedy Pixel Detector Readout), which can support a maximum acquisition rate of approximately 10 million events per second

Methods

The X-ray spectrum is firstly acquired without the sample (flat field acquisition) in order to obtain the I_0 signal for further analysis, then the sample is inserted and the I spectrum is acquired. The applied calibration parameters were derived using a combination of synchrotron radiation (energy calibration up to 38 keV), fluorescence photons and test pulses (calibration with electric signals) but high-energies calibration still needs to be refined.

A Timepix4 acquisition in data-driven mode allows to obtain the position (x, y) of each occurring event, the Time of Arrival and the Time over Threshold (ToA, ToT) of events with a fixed threshold of 700 e^- . Information gained through position, ToT and ToA is combined to perform clustering of detected events, i.e. to group hits in clusters on pixels, and to measure the energy of incident radiation. For the purposes of this work, energy information on counts and clustering of events are only taken into consideration.

Due to the charge sharing phenomenon in the considered energy range, most of the events are associated to a cluster size of 2 (i.e. the majority of incident photons involves detection by two adjacent pixels of Timepix4); therefore, K-edge analysis is firstly applied to data extracted from selection of cluster size 2, which is expected to be the most accurate for this work's purposes.

For each 100 s-lasting run, acquired energy information on counts is clustered and processed to obtain six different sets of 150 subsequent stacked images (TIFF format), corresponding to energy slicing in the range $[0 \text{ keV}, 150 \text{ keV}]$ (the actual signals extending up to 120 keV) with an energy bin width of 1 keV. Each different set of stacks is associated to a fixed cluster size, respectively $\{1, 2, 3, 4, 5, 6\}$.

Preliminary image processing

Using the ImageJ software, stacks of images belonging to the same cluster size are added: 10 TIFF files for each cluster size (from 1 to 6), for each signal (flat field and radiation transmitted through the sample) are superimposed, so to reproduce the whole acquisition statistics (counts detected during 1000 s of acquisition). The six final files for I_0 can hence be compared to the respective six final files for I , so to derive total attenuation spectra and extrapolate information about the expected lead's K-edge. Therefore, for each cluster size, stacks are processed in a final $-\ln(I/I_0)$ spectrum.

K-edge subtraction analysis

To enhance K-edge visibility, subtraction analyses are performed on Lead Carbonate's total attenuation clustered data. Having selected an energy bin width of 1 keV but being the experimental resolution characterised at least by a $FWHM = 6\text{ keV}$, total attenuation spectra are also over-stacked (summed up) to increase the actual bin width (hence, counts statistics) and potentially improve the quality of differential Kes results.

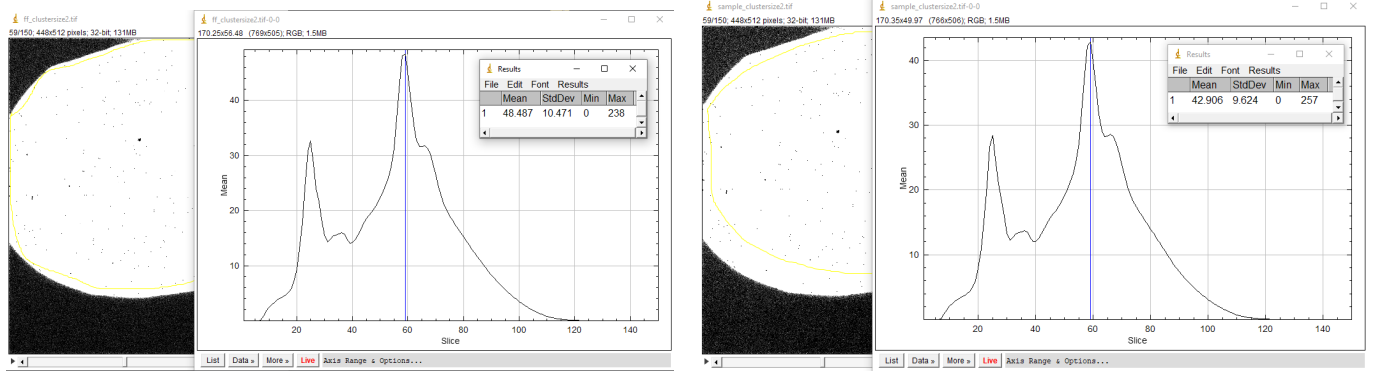
K-edge subtraction analyses are not applied to images of cluster sizes 5 and 6 because of a lack of visibility (extremely low signal to noise ratio on average) and a reduced effective energy range (energy spectra with cluster sizes 5 and 6 only extend up to $\sim 60\text{ keV}$).

On images of cluster sizes 2 and 3, which display the best visibility and the highest statistics, K-edge subtraction analyses are performed with energy bin width $\{1, 3, 4, 5\}\text{ keV}$, whereas images of cluster size 4 are analysed with energy bin width of $\{1, 3, 4\}\text{ keV}$ and images of cluster size 1 (that show the worst interpretability) are analysed with an energy bin width of 1 keV and of 3 keV .

Results

Results are reported following a descending order of visibility and calibration accuracy: analyses on stacks with cluster size 2 are used to verify the previously estimated pigment and wood thicknesses, attenuation data with cluster sizes 5 and 6 are not reported, as they turn out be unsuitable to K-edge subtraction analyses and their energy range does not reach the interested region.

Cluster size 2

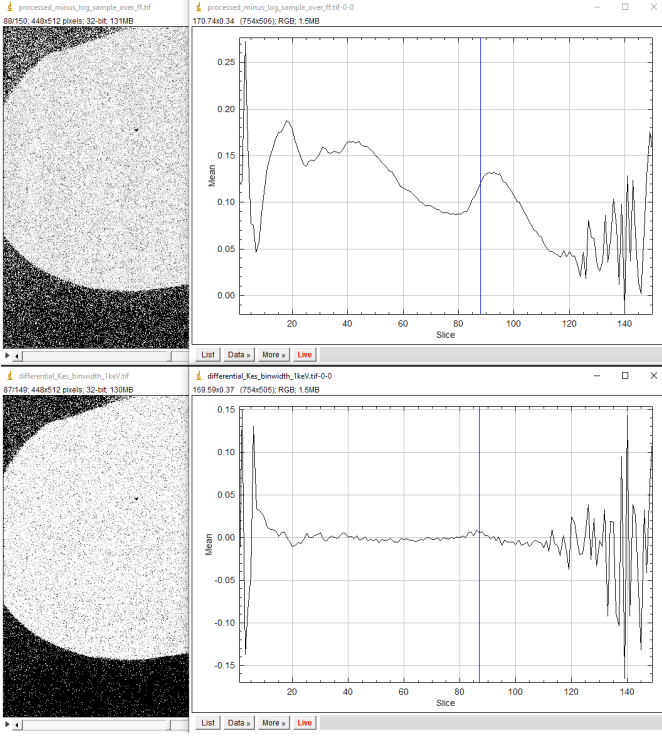


(a) Flat field counts (cluster size 2), $SNR = 4.63$

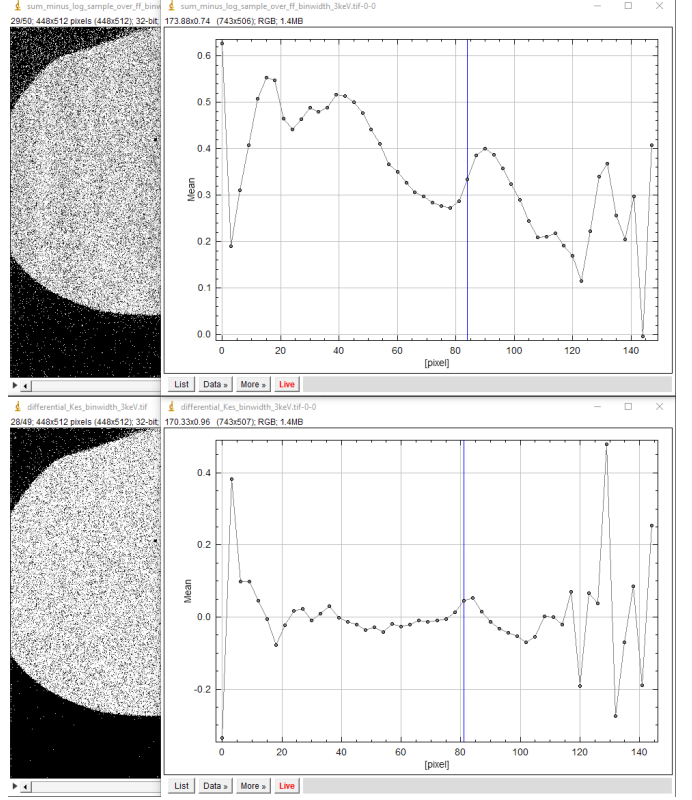
(b) Sample counts (cluster size 2), $SNR = 4.46$

Figure 2: Counts on a ROI outlined by the collimator at $E = 59\text{ keV}$ (tungsten's $K\text{-}\alpha$ fluorescence peak)

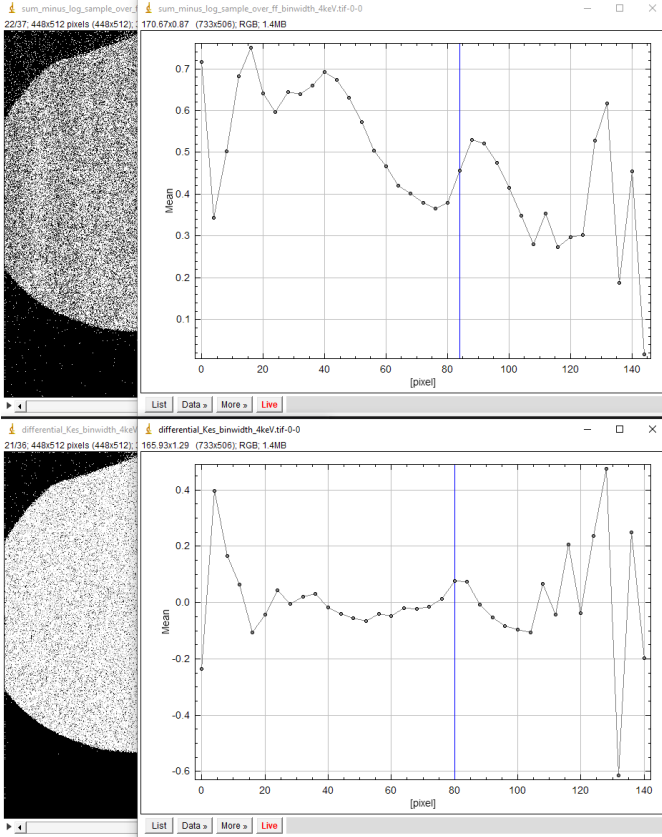
Results of total attenuation and Kes analyses show that lead's K-edge is recognisable in all four cases and its energy position is compatible with the expected one ($E_K = 88.005\text{ keV}$). K-edge subtraction analysis does not seem to be particularly useful in the case of 1 keV -bin width (such low signal might be due to oversampling and low statistics), whereas it can confirm the presence of lead when setting larger bin widths (more reasonable values if compared to the expected experimental resolution).



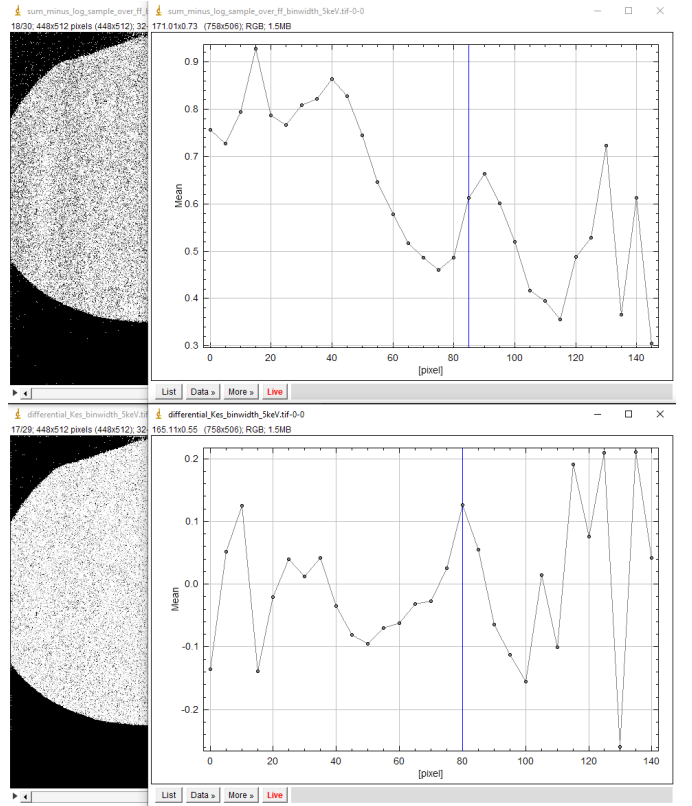
(a) Attenuation and Kes image with bin width 1 keV, slice 88



(b) Attenuation and Kes image with bin width 3 keV, slice 29



(c) Attenuation and Kes image with bin width 4 keV, slice 22



(d) Attenuation and Kes image with bin width 5 keV, slice 18

Figure 3: Total attenuation and K-edge subtraction for cluster size 2, showing lead's K-edge at $E \simeq 88$ keV

Thickness verification

Averaging over a spectral region centred in lead's K-edge (in the case of Lead Carbonate thickness estimation) and in tungsten's fluorescence peak (in the case of wood thickness estimation) and taking standard deviations as uncertainties, expected thicknesses are, respectively:

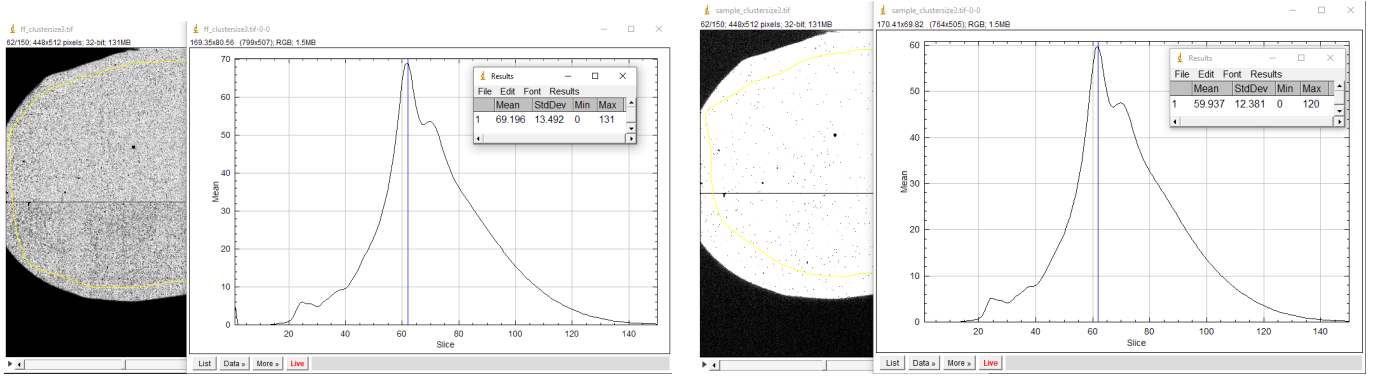
- wood: $5.04 \pm 0.52 \text{ mm}$
- Lead Carbonate: $10.10 \pm 0.30 \text{ }\mu\text{m}$

Expected total attenuation at $E_K = 88.005 \text{ keV}$ can then be computed:

$$-\ln(I/I_0) = \mu_{\text{wood}}(E_K) \cdot w_{\text{wood}} + \mu_{\text{Pb}}(E_K) \cdot w_{\text{PbCO}_3} = (1.38 \pm 0.08) \cdot 10^{-1}$$

Total attenuation at slice 88 and 89, i.e. around $E = 88 \text{ keV}$, can be measured on the processed TIFF file associated to a cluster size of 2: using the *measure* function on the software ImageJ yields $-\ln(I/I_0) = (1.23 \pm 2.41) \cdot 10^{-1}$. Despite the large uncertainty on the measured attenuation (which might be due to a relatively low counts signal to be processed), the two resulting attenuation values can be assumed to be compatible. The previously estimated thicknesses are therefore validated.

Cluster size 3

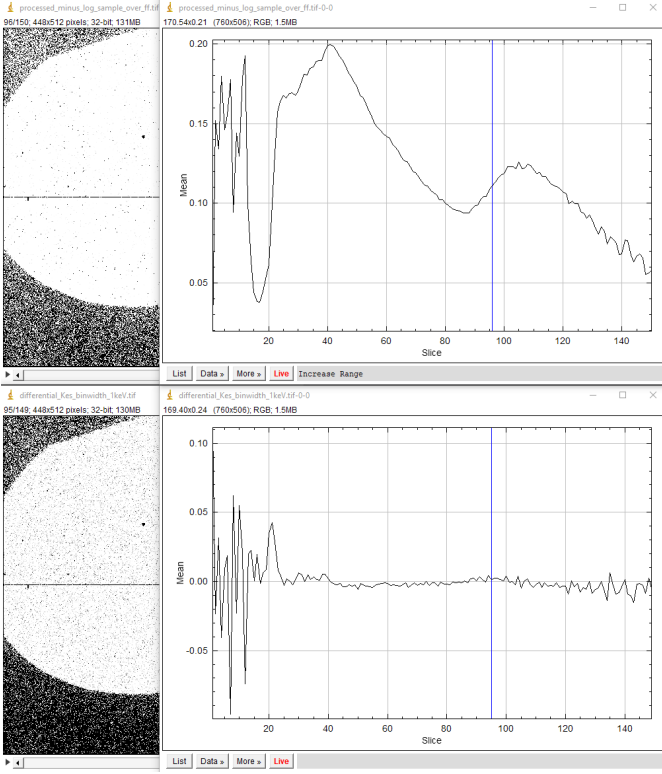


(a) Flat field counts (cluster size 3), $SNR = 5.13$

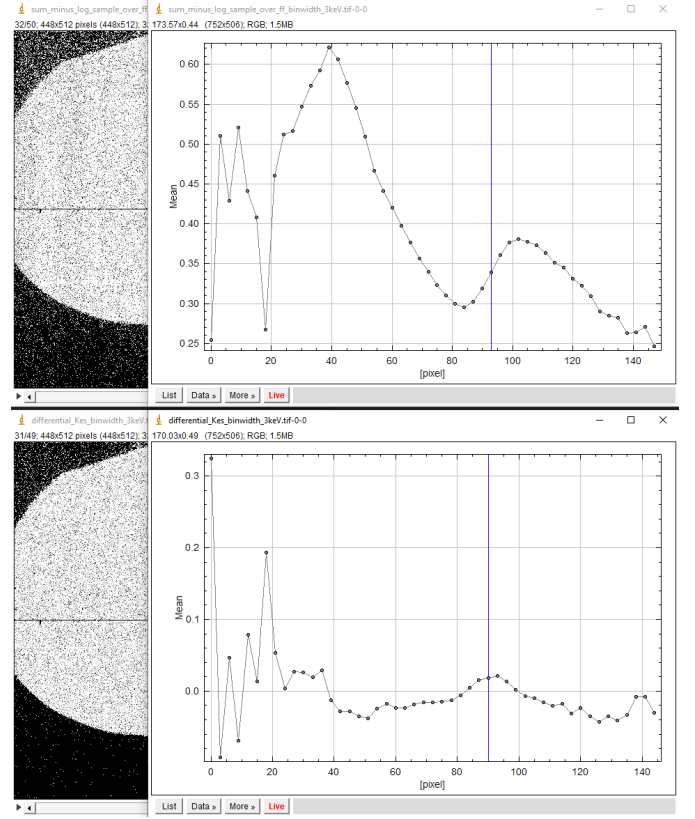
(b) Sample counts (cluster size 3), $SNR = 4.84$

Figure 4: Counts on a ROI outlined by the collimator at $E = 62 \text{ keV}$ (tungsten's K- α peak, suboptimal calibration)

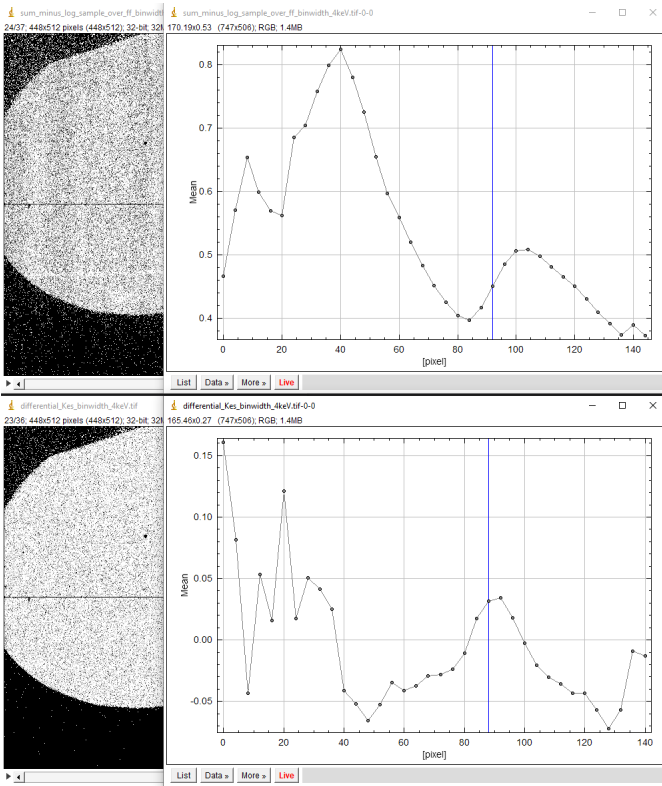
Results of total attenuation and Kes analyses show that lead's K-edge can be recognised, but calibration corrections are required (as it can be seen on counts of cluster size 3). Similarly to the case of cluster size 2, with cluster size 3 K-edge subtraction analysis does not seem to be particularly useful in the case of 1 keV -bin width, but it could confirm the presence of lead if setting larger bin widths.



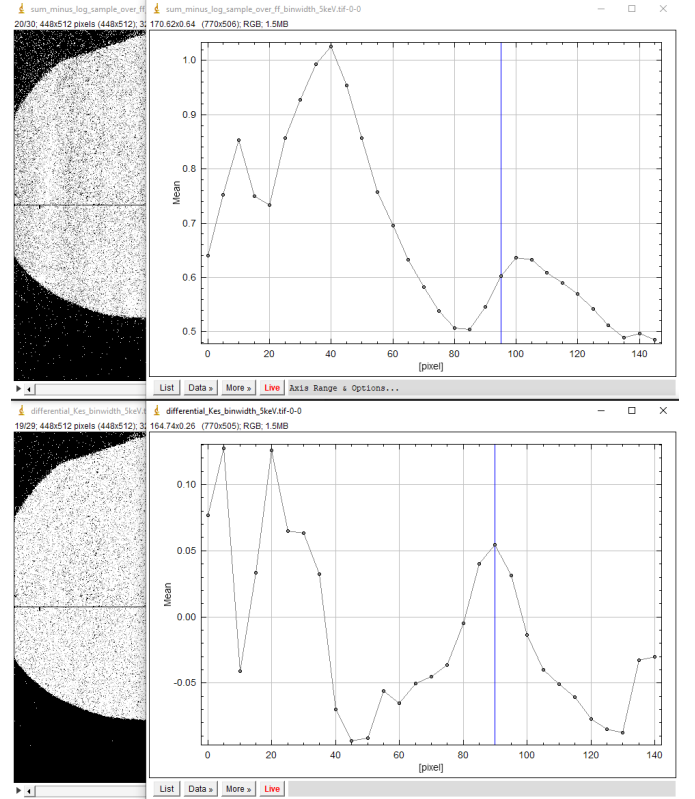
(a) Attenuation and Kes image with bin width 1 keV , slice 96



(b) Attenuation and Kes image with bin width 3 keV , slice 32



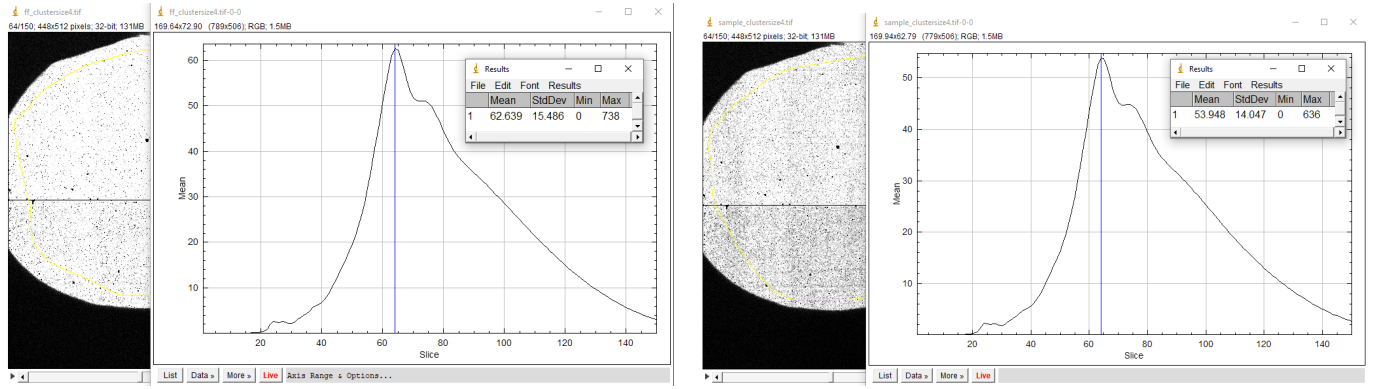
(c) Attenuation and Kes image with bin width 4 keV , slice 24



(d) Attenuation and Kes image with bin width 5 keV , slice 20

Figure 5: Total attenuation and K-edge subtraction for cluster size 3, showing a K-edge at $E \simeq 96\text{ keV}$

Cluster size 4

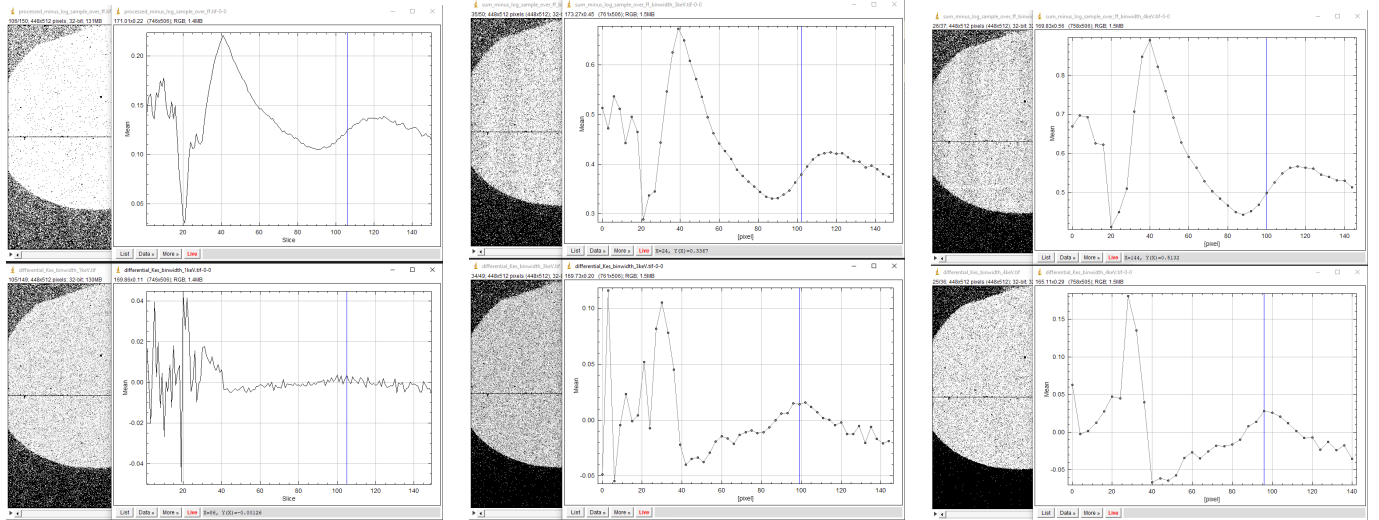


(a) Flat field counts (cluster size 4), $SNR = 4.05$

(b) Sample counts (cluster size 4), $SNR = 4.27$

Figure 6: Counts on a ROI outlined by the collimator at $E = 64 \text{ keV}$ (tungsten's $K\alpha$ peak, suboptimal calibration)

Results of total attenuation and Kes analyses show that lead's K-edge can be recognised, but strong calibration corrections are required (as it can be seen on counts of cluster size 4). Similarly to the case of cluster sizes 2 and 3, with cluster size 4 K-edge subtraction analysis does not seem to be particularly useful in the case of 1 keV -bin width, but it seems to be enough to confirm the presence of lead if a bin width of 4 keV is set.



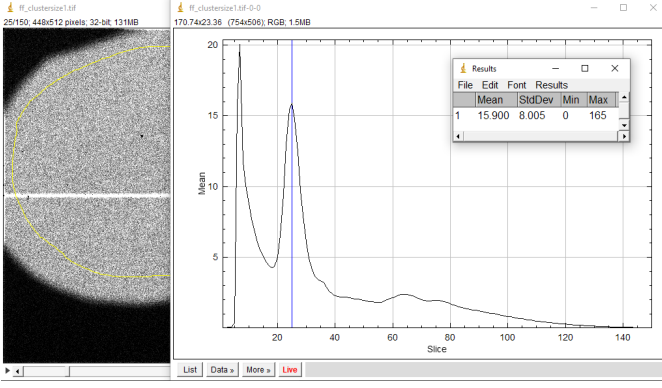
(a) Attenuation and Kes image with bin width 1 keV at slice 106

(b) Attenuation and Kes image with bin width 3 keV at slice 35

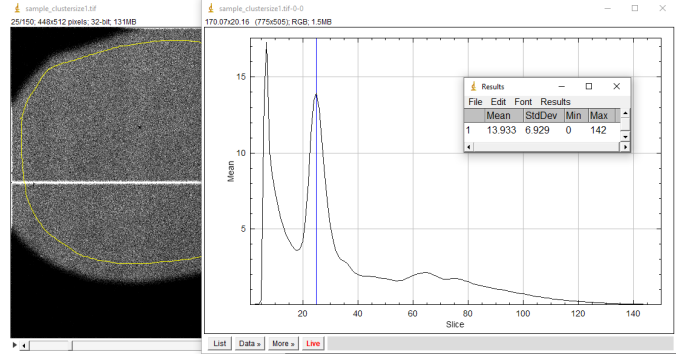
(c) Attenuation and Kes image with bin width 4 keV at slice 26

Figure 7: Total attenuation and K-edge subtraction for cluster size 4, showing a K-edge at $E \simeq 105 \text{ keV}$

Cluster size 1



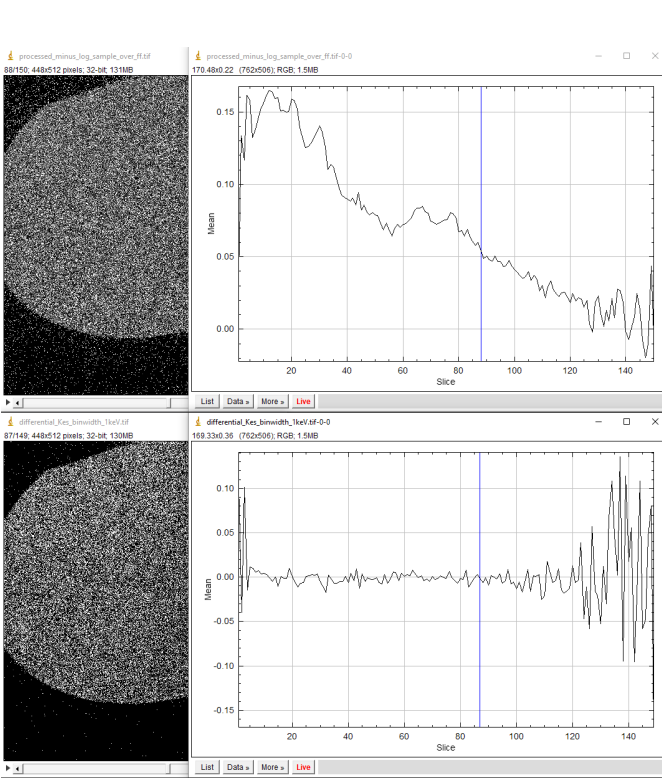
(a) Flat field counts (cluster size 1), $SNR = 1.99$



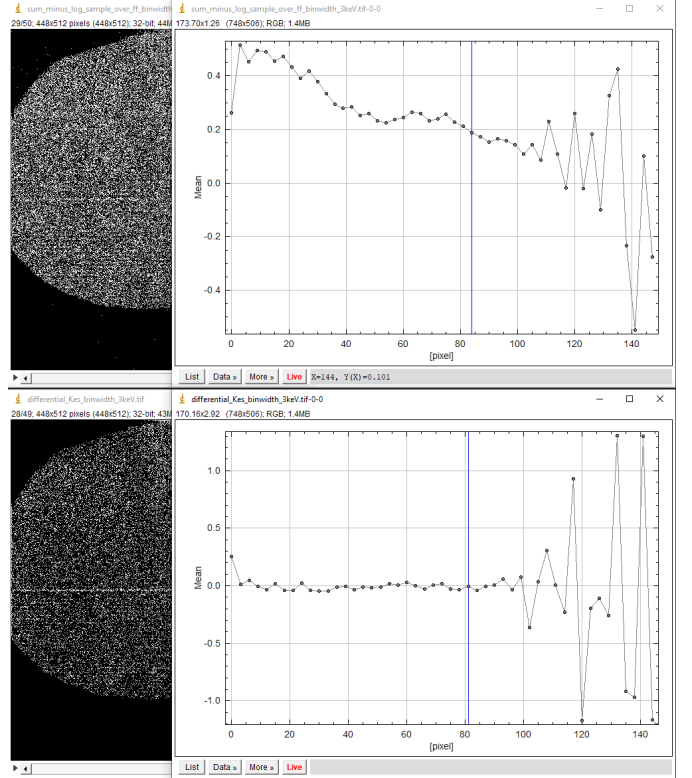
(b) Sample counts (cluster size 1), $SNR = 2.01$

Figure 8: Counts on a ROI outlined by the collimator at $E = 25 \text{ keV}$ (cadmium's K- α fluorescence peak)

Results of total attenuation and Kes analyses show no presence of a K-edge: because of the relatively high energy of lead's K-edge, it can be assumed that all photons associated to such energy values cause a charge sharing effect on at least two adjacent pixels; therefore, events with cluster size 1 are not expected to include the ones attributable to lead transmission at high energies. As a matter of fact, counts of events with cluster size 1 only display a strong cadmium fluorescence peak (because of the employed detector), whereas peaks expected at higher energies (e.g. tungsten fluorescence peak at $E = 59.318 \text{ keV}$) are almost negligible.



(a) Attenuation and Kes image with bin width 1 keV , slice 88



(b) Attenuation and Kes image with bin width 3 keV , slice 29

Figure 9: Total attenuation and K-edge subtraction for cluster size 1, unable to show lead's K-edge at $E \simeq 88 \text{ keV}$

Cluster sizes 5 and 6

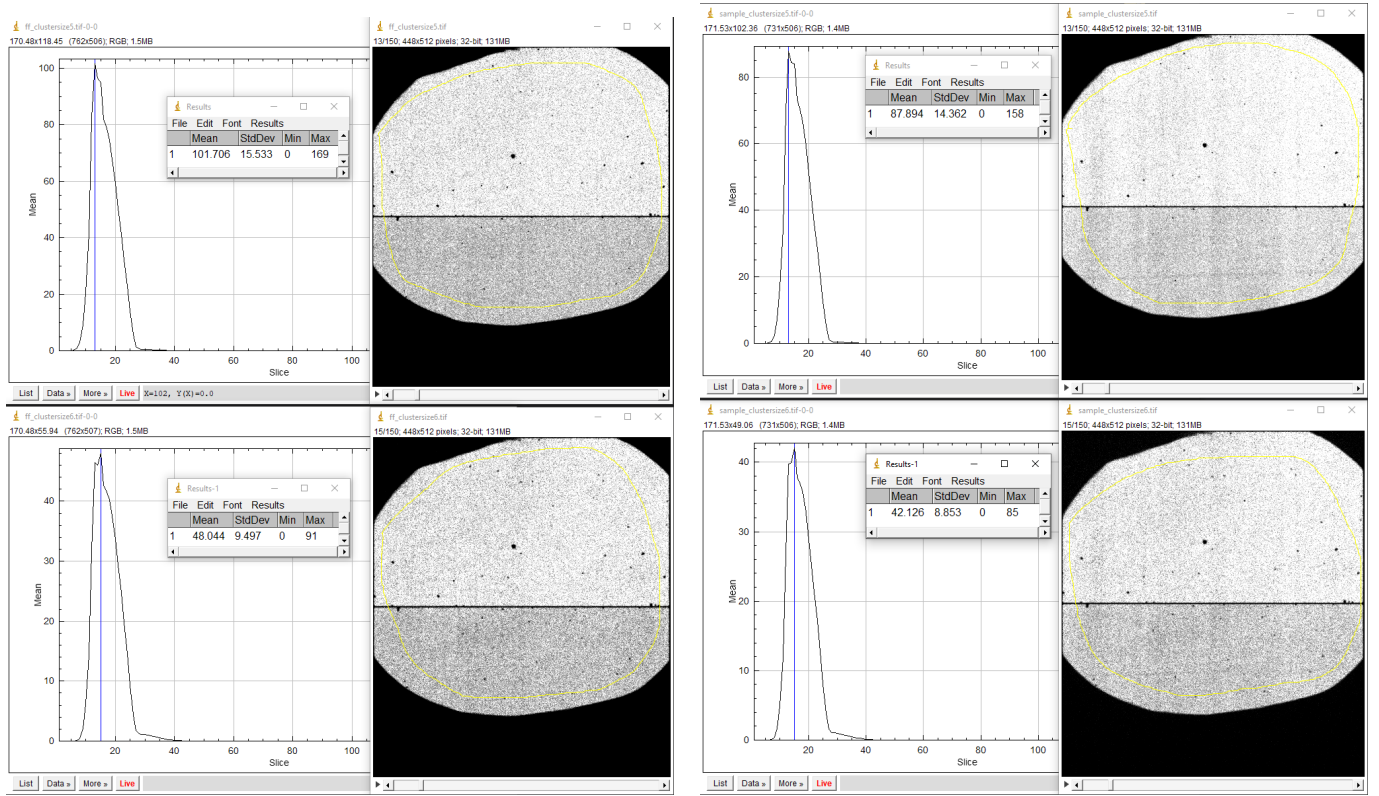


Figure 10: Counts on a ROI outlined by the collimator at $E = 13 \text{ keV}$ (cluster size 5) and $E = 15 \text{ keV}$ (cluster size 6)

No further analyses are performed on data with cluster size 5 and cluster size 6.

Conclusions

X-ray analyses on a Lead Carbonate sample with Timepix4 seem to prove that such technique can be used to identify lead-based pigments using K-edge analysis, as long as the pigment has a thickness of at least $\sim 10 \mu\text{m}$ and accurate calibration parameters are applied. In particular, a cluster size of 2 seems to be the most suitable to analyse attenuation in the considered energy range.

Links

- [1] Timepix4 specifications [here](#)
- [2] Tables of linear attenuation coefficients and material densities from NIST [here](#)
- [3] Material K-edges can be found in the X-ray data booklet [here](#)
- [4] Details about previous analyses [here](#)